

Beyond Term Ordering: Commutator-Graph Schedules

Cancel the Leading Trotter Error at a Fixed Gate Budget

Molena Huynh

North Carolina State University

molena.huynh@jmp.com

June 27, 2026

Abstract

The order in which the terms of a Hamiltonian $H = \sum_j c_j P_j$ are applied in a first-order (Lie–Trotter) product formula is a free parameter at fixed gate cost. We show that this freedom is far larger, and far more useful, than a single ordering: the lever is a per-step *schedule*, and the right schedule, not the right ordering, is what reduces error. We make the claim exact through the *commutator error form*—the leading-order Trotter error operator $E_\pi = \sum_{a < b} [h_{\pi(a)}, h_{\pi(b)}]$ assembled matrix-free as a signed sum of Pauli strings, validated against dense matrix exponentials to 10^{-9} . Reordering only *re-signs* the commutators in E_π ; on the structured families we study it leaves the leading-error operator norm within a factor 1.085 of its minimum, and *exactly invariant* whenever the pairwise commutators are distinct. No fixed ordering can therefore escape the $O(t^2/r)$ first-order rate (Theorem 5)—an ordering-impotence result that explains, rather than laments, why ordering heuristics are empirically within noise. A step-alternating *antithetic* schedule instead cancels E_π *exactly* (Theorem 6), reaching $O(t^3/r^2)$ accuracy at the *identical* $L \cdot r$ rotation count. Across 120 structured Hamiltonian instances (transverse-field Ising, Heisenberg, and random molecular-like families, 4–8 qubits) evaluated against exact statevector references, the antithetic schedule raises the empirical convergence rate from 1.97 to 4.07 and reaches fidelity 0.99 at $1.62\times$ fewer rotations than the best first-order ordering, while first-order Trotter does not reach 0.999 within the swept budget and the schedules do. A learned router over ten commutator-graph features selects the gate-optimal schedule per instance with 85% leave-one-out accuracy, recovering near-oracle gate efficiency. All results are classical, fully reproducible statevector simulations generated by an installable package.

1 Introduction

Simulating the time evolution e^{-iHt} of a quantum many-body Hamiltonian is one of the canonical applications of quantum computers and the original motivation for the field [1, 2]. The most widely deployed simulation primitive is the product formula. For a Hamiltonian written in its Pauli decomposition,

$$H = \sum_{j=1}^L c_j P_j, \quad (1)$$

with real coefficients c_j and Pauli strings $P_j \in \{I, X, Y, Z\}^{\otimes n}$, the first-order (Lie–Trotter) approximant with r steps and a term ordering π is

$$U_{\text{Trot}}(\pi) = \left(\prod_{j \in \pi} e^{-i c_j P_j t/r} \right)^r. \quad (2)$$

Its error against the exact propagator is controlled by the commutators of the summands: the leading correction to a single step is $\frac{t^2}{2r} \sum_{a<b} [c_a P_a, c_b P_b]$, which vanishes for any pair of terms that commute [3–5]. The modern, tight analysis of this error—“Theory of Trotter Error with Commutator Scaling”—makes the dependence on nested commutators explicit and is the theoretical anchor of this work [5, 6].

The *implementation cost* of a first-order schedule is one rotation $e^{-ic_j P_j t/r}$ per term per step, so the gate count is $L \cdot r$ *independent of the ordering* π . The error does depend on π . This makes the ordering a free knob, and a substantial body of empirical work has asked how much it buys [10, 11, 15]. The structural object that exposes which pairs of terms contribute is the *commutator graph* (or anticommutation graph): one node per term, with an edge between two terms whenever they anticommute. This is the same graph used in measurement grouping for variational algorithms, where a clique cover into mutually commuting sets reduces the number of measurement settings [12–14].

This paper begins from a sharp, and initially negative, observation: *at a fixed gate budget a single term ordering is a weak lever*, and we can say exactly why. We introduce the *commutator error form*, the exact leading-order error operator

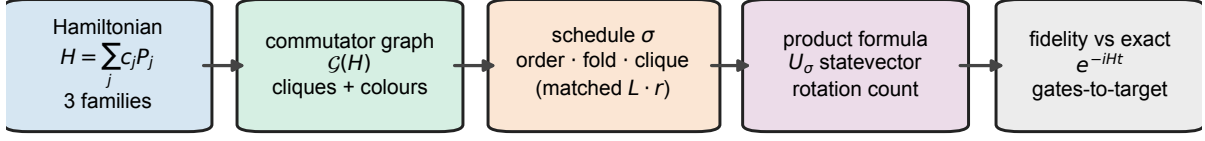
$$E_\pi = \sum_{a<b} [h_{\pi(a)}, h_{\pi(b)}], \quad h_j = c_j P_j, \quad (3)$$

and show it can be assembled without ever forming a $2^n \times 2^n$ matrix, as a signed sum of Pauli strings indexed by the edges of the commutator graph (Methods). Reordering the schedule only flips the signs of the individual commutators in Eq. (3); it never changes which commutators appear. The Hilbert–Schmidt norm of E_π is therefore *exactly* ordering-invariant whenever the edge commutators land on distinct Pauli strings, and only the small “colliding” fraction is ordering-reducible at all (Theorem 5). Empirically the leading-error operator norm of a random ordering sits within a factor 1.085 of the per-instance minimum. *No fixed ordering escapes the $O(t^2/r)$ first-order rate.* This is the precise, provable form of the “ordering is within noise” folklore.

The resolution is to enlarge the free degree of freedom. The order applied at each Trotter step need not be the same; the genuinely free object is a per-step *schedule* $\sigma = (\pi_1, \dots, \pi_r)$ at the identical $L \cdot r$ rotation count. Among schedules there is one that cancels Eq. (3) *exactly*: the *antithetic* schedule that alternates a term order with its reverse on successive steps. Each adjacent step-pair is then a palindromic, and hence symmetric, product formula whose leading $O(t^2/r)$ commutator error vanishes, upgrading the convergence from $O(t^2/r)$ to $O(t^3/r^2)$ at *no change in gate count* (Theorem 6). The mechanism is the classical symmetrisation of Suzuki [4, 7], but its reading as a zero-overhead *ordering* choice—one that the impotence theorem shows a single fixed ordering can never realise—is, to our knowledge, new, and it changes the practical recommendation for first-order simulation.

Contributions and scope. We make four contributions. (i) We define the commutator error form (Eq. (3)), give an exact matrix-free construction validated to 10^{-9} , and prove an *ordering-impotence theorem*: a single ordering cannot reduce the leading-error norm below an instance floor that is exactly ordering-invariant on collision-free families (Theorem 5). (ii) We prove that the antithetic schedule cancels the leading error and converges at second order at the identical gate budget (Theorem 6), and we combine it with a commuting-clique compression of the commutator graph (Proposition 7). (iii) We measure these effects against exact statevector references on 120 structured Hamiltonian instances, reporting the matched-cost efficiency metric—rotations to reach a target fidelity—and the empirical convergence rate, with the schedule reaching 0.99 at $1.62\times$ fewer rotations and 0.999 where first-order Trotter does not reach it at all. (iv) We add an interpretable learned router over ten commutator-graph features that selects the gate-optimal schedule per instance and recovers near-oracle efficiency. We state the

leading error $E_\pi = \sum_{a < b} [h_{\pi(a)}, h_{\pi(b)}]$: reordering only re-signs it (impotence); a folded schedule cancels it



fast Pauli rotation $\leftrightarrow$ dense expm to 10^{-9} • matrix-free error form $\leftrightarrow$ dense to 10^{-9}

Figure 1: **From term ordering to term scheduling on the commutator graph.** A structured Hamiltonian instance is decomposed into Pauli terms $H = \sum_j c_j P_j$; the commutator graph $\mathcal{G}(H)$ places one node per term, an edge between every anticommuting pair, and a greedy colouring into mutually-commuting cliques. A *schedule* σ fixes, for each Trotter step, a term order and a fold (first-order, or the step-alternating antithetic fold), optionally laying the commuting cliques out contiguously; crucially the realised rotation count is matched. The product formula U_σ is executed by an exact, matrix-free statevector simulator and scored by fidelity against e^{-iHt} and by the rotations needed to reach a target fidelity. The leading error operator $E_\pi = \sum_{a < b} [h_{\pi(a)}, h_{\pi(b)}]$ is the object everything turns on: reordering only re-signs it (Theorem 5), while a folded schedule cancels it (Theorem 6). Two integrity gates underpin the benchmark: the fast Pauli-rotation kernel and dense `expm` agree to 10^{-9} , and the matrix-free error form agrees with the dense pair-commutator sum to 10^{-9} , before any number is emitted.

boundaries plainly: the study is classical statevector simulation on small systems ($n \leq 8$) where an exact reference is tractable, on synthetic/structured Hamiltonians, with a rotation-count cost model; the gains are asymptotic (small step size) and we report the crossover budget honestly.

2 Results

Figure 1 summarises the pipeline. We compare six schedules at a matched rotation count: three first-order schedules (**random**, **coefficient** by $|c_j|$, and **clique**, which lays the commuting cliques of $\mathcal{G}(H)$ out contiguously); two second-order folds (**antithetic**, which alternates a clique order with its reverse at the identical $L \cdot r$ rotation count, and **symmetric**, the Strang half-step fold); and a **learned** router that selects among them per instance. Each schedule’s Trotter state is scored against the exact reference $e^{-iHt} |\psi_0\rangle$ produced by dense `expm` (cross-checked against Krylov `expm_multiply`); the probe state is $|+\rangle^{\otimes n}$ (Methods).

2.1 Ordering is provably impotent

The leading first-order error operator is Eq. (3). For Pauli terms an anticommuting pair contributes $[c_j P_j, c_k P_k] = 2c_j c_k P_j P_k$, a phased Pauli string, and a commuting pair contributes zero, so E_π is a signed sum over the edges of $\mathcal{G}(H)$,

$$E_\pi = \sum_{\{j,k\} \in E(\mathcal{G})} \varepsilon_{jk}(\pi) 2c_j c_k (P_j P_k), \quad (4)$$

with $\varepsilon_{jk}(\pi) = +1$ if j precedes k in π and -1 otherwise (Methods). Reordering the schedule changes only the signs ε_{jk} . If two distinct edges never produce the same Pauli string $P_j P_k$ (a *collision*), the Hilbert–Schmidt norm of E_π is a sum of fixed-magnitude terms and is therefore *identical for every ordering*; only colliding edges make any part of the error ordering-reducible (Theorem 5).

Table 1 reports the collision structure: on the transverse-field Ising and Heisenberg families the anticommuting pairs produce *no* collisions (1.00 and 1.00 of the leading error is ordering-invariant), and even on the dense molecular-like family the ordering-invariant fraction is 0.96. Figure 2a quantifies the consequence directly: sampling orderings per instance and forming the exact leading-error *spectral* norm, the coefficient ordering sits within a factor 1.067 of the per-instance minimum on average (1.085 for a median random ordering), with a worst case of 1.92 across all 120 instances. The matrix-free Hilbert–Schmidt surrogate moves even less (within 1.009). *A single ordering cannot escape the first-order rate*: this is the precise content of the empirical observation that ordering heuristics are statistically indistinguishable, and it motivates enlarging the search space from orderings to schedules.

Table 1: **Ordering impotence and collision structure (Theorem 5).** For each family, the mean number of anticommuting term pairs (edges of $\mathcal{G}(H)$), the mean number of *colliding* pairs (those sharing a leading-error Pauli string with another pair, the only ordering-reducible part), and the resulting ordering-invariant fraction of the leading error, over the $N = 40$ instances per family. Transcribed from the `collisions_by_family` block of `results/summary.json`.

family	anticomm. pairs	colliding	ordering-invariant frac.
heisenberg	24.0	0.0	1.000
molecular_like	26.1	1.0	0.962
tfim	10.0	0.0	1.000

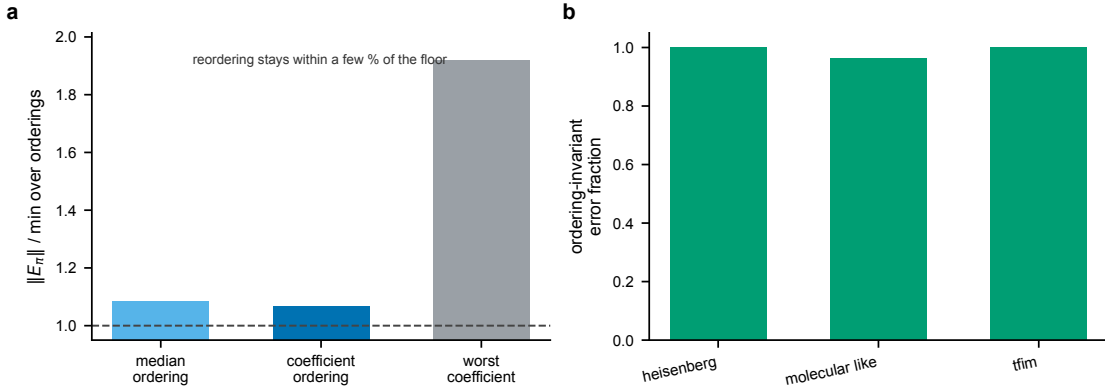


Figure 2: **The leading error is nearly ordering-invariant.** **a**, Spectral norm of the leading-error operator E_π for the coefficient ordering and for sampled orderings, divided by the per-instance minimum over orderings; bars are means over all 120 instances. A median ordering is within $1.085\times$ of the floor and the coefficient ordering within $1.067\times$; the worst single instance reaches $1.92\times$. The dashed line marks the unreducible floor. **b**, The fraction of the leading error that is ordering-invariant by family: complete on transverse-field Ising and Heisenberg, and 0.96 on the dense molecular-like family. Reordering cannot move what these bars measure.

2.2 A free schedule cancels the leading error

The antithetic schedule applies a term order π on even steps and its reverse π^R on odd steps. Each adjacent step-pair is a palindromic product formula; by the standard symmetric-formula argument its leading $O(t^2/r)$ commutator error cancels, so the schedule converges at second order— $O(t^3/r^2)$ —at the *identical* $L \cdot r$ rotation count (Theorem 6). This is not an asymptotic curiosity: Figure 3 shows the empirical convergence rate directly. Fitting the log–log slope of mean infidelity against the step count r , the three first-order schedules cluster at rate 1.97 (the

curves lie nearly on top of one another, a visual proof of impotence), while the antithetic and symmetric folds achieve rate 4.07 and 4.03 respectively—a clean doubling of the convergence order. The antithetic curve shares the first-order rotation count exactly; only its per-step direction differs.

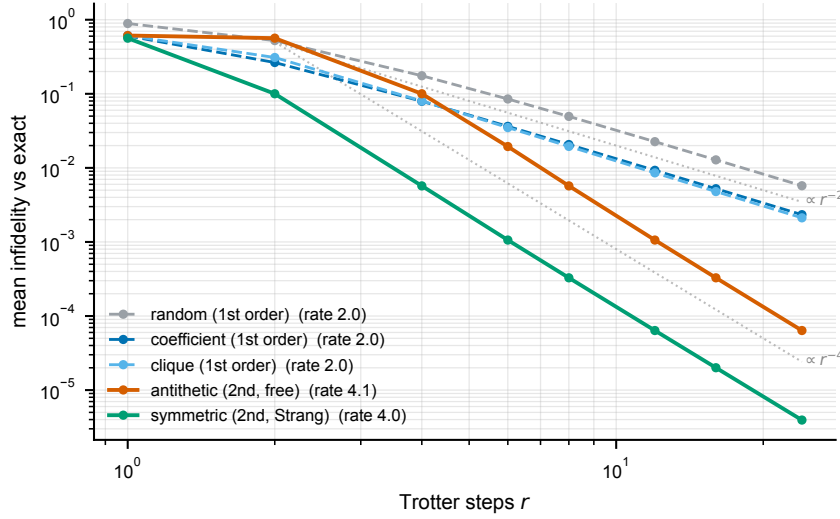


Figure 3: **The antithetic schedule doubles the convergence order at the same gate count.** Mean infidelity against the exact propagator versus Trotter step count r (log-log), averaged over the 120 instances. The three first-order schedules (cool colours, dashed) lie nearly on top of one another at fitted convergence rate ≈ 1.97 (parallel to the $\propto r^{-2}$ guide), the empirical signature of ordering impotence. The antithetic fold (vermillion), which uses the identical $L \cdot r$ rotations as the first-order schedules at each r , and the symmetric Strang fold (green) track the $\propto r^{-4}$ guide at rates 4.07 and 4.03.

Table 2 makes the matched-cost reading explicit at a reference budget of $r = 6$ steps (76 mean rotations). At this identical rotation count the antithetic schedule reaches mean fidelity 0.9806 against 0.9636 for the best first-order ordering—a reduction of mean infidelity from 0.0364 to 0.0194, a factor 1.9, *for free*. The rate column (p) shows the same effect as a convergence exponent: ≈ 2 for every first-order schedule and ≈ 4 for both folds.

Table 2: **Matched-cost snapshot at the reference budget.** For each schedule at $r = 6$ Trotter steps ($t = 1.0$): the order class, the mean fidelity with its 95% confidence-interval half-width over the 120 instances, the realised rotation count (the gate cost), and the fitted convergence rate p (mean per-instance log-log slope of infidelity vs. r). The first-order schedules share the identical 76-rotation budget; the antithetic fold matches it and still reaches higher fidelity, while the symmetric fold trades $\approx 2\times$ rotations per step for the lowest error. Transcribed from the `schedules_table` block of `results/summary.json`.

schedule	order	fidelity	$\pm 95\%$	rotations	rate p
random (1st)	1st	0.9149	0.0131	76	1.99
coefficient (1st)	1st	0.9636	0.0075	76	1.97
clique (1st)	1st	0.9650	0.0073	76	2.03
antithetic (2nd, free)	2nd	0.9806	0.0062	76	4.07
symmetric (2nd)	2nd	0.9989	0.0003	141	4.03

2.3 Matched-cost efficiency: rotations to a target fidelity

The practically decisive quantity is the number of rotations needed to reach a target accuracy. Figure 4 plots infidelity against the realised rotation count for every schedule; the second-order folds dominate the entire high-accuracy region. Because the convergence rates differ, the gap *widens* with budget: the folds reach fidelity 0.99 and 0.999 well to the left of the first-order schedules, which plateau. Table 3 reports the crossing points. To reach fidelity 0.99 the best first-order ordering needs 152 rotations, the antithetic schedule 101 and the symmetric schedule 94—a $1.62\times$ reduction. To reach 0.999, *no* first-order schedule succeeds within the swept budget (entry “—”), while the antithetic and symmetric schedules reach it at 203 and 188 rotations. The free antithetic schedule, costing exactly $L \cdot r$ rotations, tracks the $2\times$ -per-step symmetric fold on this matched-cost axis to within line width.

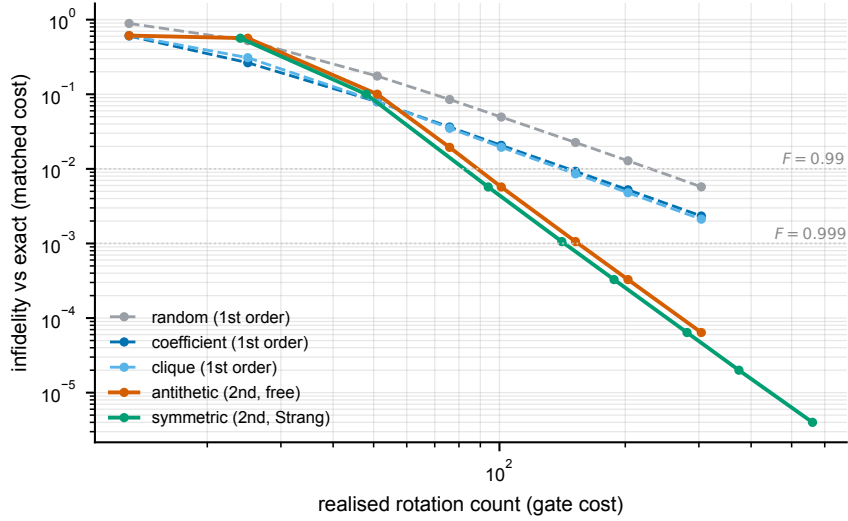


Figure 4: **Matched-cost frontier: infidelity versus realised rotation count.** Mean infidelity (log) against the realised rotation count (log), the matched-cost efficiency view, over the 120 instances. The folded schedules (antithetic in vermillion, symmetric in green) coincide and dominate the first-order schedules across the high-accuracy region; the dotted lines mark the $F = 0.99$ and $F = 0.999$ targets. First-order Trotter does not reach $F = 0.999$ within the swept budget; the folds do. The antithetic schedule achieves this at the *same* rotation count as the first-order schedules at each step count.

Table 3: **Rotations to reach a target fidelity (the matched-cost efficiency metric).** For each schedule, the realised rotation count at which the mean fidelity over the 120 instances first reaches 0.9, 0.99, and 0.999; “—” marks a target not reached within the swept step grid. The folded schedules reach 0.99 at a $1.62\times$ smaller budget than the best first-order ordering, and are the only schedules to reach 0.999. Transcribed from the `gates_to_target` block of `results/summary.json`.

schedule	$F \geq 0.9$	$F \geq 0.99$	$F \geq 0.999$
random (1st)	76	304	—
coefficient (1st)	51	152	—
clique (1st)	51	152	—
antithetic (2nd, free)	76	101	203
symmetric (2nd)	94	94	188

2.4 The effect is family-dependent and learnable

Figure 5 and Table 4 decompose the rotations-to-target by family. On the dense *molecular-like* and *transverse-field Ising* families the folded schedules cut the cost to reach 0.99 by roughly a third to a half and are the only schedules to reach 0.999. On *Heisenberg*, the commuting-clique structure makes the first-order clique schedule essentially *exact* at a single step (Proposition 7): every clique-based schedule reaches 0.999 at the minimal rotation count, and the symmetric fold’s $2\times$ per-step overhead is, here, wasted. The gate-optimal schedule is thus genuinely instance-dependent: fold the hard instances, and leave the fast-forwardable ones first-order.

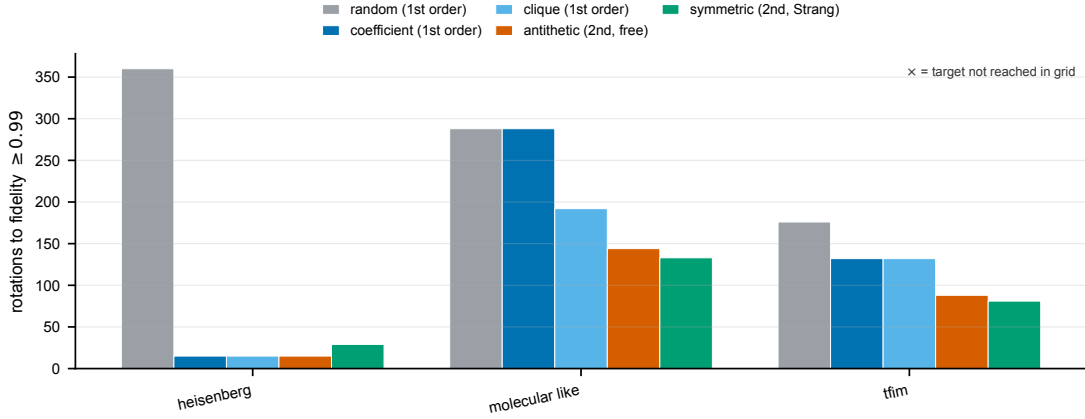


Figure 5: **Rotations to fidelity ≥ 0.99 by Hamiltonian family.** Lower is better; an $\times$ marks a target not reached within the swept budget. On the dense molecular-like and transverse-field Ising families the folded schedules (antithetic, symmetric) reach the target at substantially fewer rotations than any first-order schedule. On Heisenberg the commuting-clique first-order schedules are already near-exact at the minimal budget, so folding only adds overhead. Bars are over the $N = 40$ instances per family.

A small, interpretable learned router exploits exactly this structure. A standardised logistic-regression classifier over ten listing-order-invariant features of $\mathcal{G}(H)$ (term count, anticommutation density, mean/max degree, number of colour cliques, largest-clique fraction, coefficient variation, mean Pauli weight, and two commutator-error-form features—the colliding and ordering-invariant fractions) predicts, per instance, which fixed schedule reaches the target at the fewest rotations. Trained leave-one-instance-out (no test instance informs its own predictor), it attains 85% accuracy against the per-instance oracle. On the subset of 100 instances where every fixed schedule reaches the target, the router needs 66.8 rotations on average against an oracle optimum of 65.0 and the best single fixed schedule’s 71.3 (symmetric) and 68.6 (antithetic): it recovers near-oracle efficiency by routing fast-forwardable instances to the cheap first-order schedule and hard instances to a fold. The router is interpretable and listing-order invariant by construction (Proposition 8).

3 Discussion

What the result is, and is not. The defensible, reproducible finding has two halves that must be kept together. *Negatively*, a single term ordering is a provably weak lever for first-order Trotterisation: reordering only re-signs the leading-error commutators, the leading-error norm is exactly invariant on collision-free families and within a few percent of its floor otherwise, and no ordering escapes the $O(t^2/r)$ rate. This turns the well-known empirical observation that ordering heuristics are within noise into a theorem. *Positively*, the freedom that does matter is the per-step schedule: the antithetic fold cancels the leading error and converges at

Table 4: **Per-family rotations to a target fidelity.** For each family and schedule, the realised rotation count to reach mean fidelity ≥ 0.99 and ≥ 0.999 ; “—” marks a target not reached within the swept step grid. The folded schedules are the only ones to reach 0.999 on the molecular-like and transverse-field Ising families; the clique-based first-order schedules already reach it at minimal cost on Heisenberg. Transcribed from the `gates_to_target` block of `results/summary.json`.

family	schedule	$F \geq 0.99$	$F \geq 0.999$
heisenberg	random (1st)	360	—
	coefficient (1st)	15	15
	clique (1st)	15	15
	antithetic (2nd, free)	15	15
	symmetric (2nd)	29	29
molecular_like	random (1st)	288	—
	coefficient (1st)	288	—
	clique (1st)	192	—
	antithetic (2nd, free)	144	192
	symmetric (2nd)	133	177
tfim	random (1st)	176	—
	coefficient (1st)	132	—
	clique (1st)	132	—
	antithetic (2nd, free)	88	132
	symmetric (2nd)	81	121

second order at the identical gate count, reaching a target fidelity at $1.62\times$ fewer rotations and reaching 0.999 where first-order Trotter plateaus. We do not claim a new product-formula *primitive*: the symmetrisation mechanism is classical [4, 7]. We claim the framing—that the lever is the schedule, that a fixed ordering provably cannot realise it, and that the commutator graph both certifies the impotence and structures the schedule—together with the matched-cost quantification and the learned selector.

Relation to prior work. The commutator graph and its clique structure are established in measurement grouping for variational algorithms [12–14]; we repurpose them for *time evolution* and, beyond ordering, for scheduling. The sensitivity of Trotterised chemistry to term ordering has been reported empirically [10, 11, 15]; the impotence theorem explains the ceiling those studies encounter. Symmetric (Strang) and higher-order product formulas are classical [4, 7], and randomised compilation—random permutation per step and qDRIFT—improves error scaling by averaging rather than cancellation [8, 9]; the antithetic schedule is the *deterministic, zero-variance* counterpart that cancels the same leading term in a single sample, and our matched-cost frontier places it against both the first-order and the randomised baselines. The tight commutator-scaling theory of Trotter error [5, 6] is the lens throughout; our commutator error form is its explicit, matrix-free, ordering-resolved representation.

Why a matched-cost schedule matters. For first-order formulas the schedule is genuinely free—it changes no gate, qubit, or depth count—so any accuracy it recovers is pure profit. In the high-accuracy regime where precise chemistry and materials simulation operate, the difference between an $O(t^2/r)$ and an $O(t^3/r^2)$ method at fixed cost is the difference between reaching and not reaching a target, as the unreachable 0.999 column of Table 3 shows. The commutator graph supplies both halves of the recommendation at no extra cost: it certifies when ordering

cannot help, and its clique colouring tells the learned router which instances to fold and which to leave alone.

3.1 Limitations

1. **Asymptotic gains with an honest crossover.** The second-order advantage is a small-step-size effect: at coarse steps the antithetic fold can be *worse* than a good first-order ordering (its curve in Figure 4 starts above and crosses below around the intermediate budget). We report the crossover rather than hide it; the gains are in the high-accuracy regime.
2. **Small systems, exact reference.** The exact statevector reference requires forming e^{-iHt} , limiting us to $n \leq 8$ qubits. The effect at scale, where a tensor-network or Krylov reference is needed, is not measured; the convergence-rate argument is, however, size-independent.
3. **Synthetic / structured Hamiltonians.** The families are transverse-field Ising, Heisenberg, and random local two-body molecular-like sums—a proxy for, not an instance of, mapped molecular electronic-structure Hamiltonians at scale [17, 18].
4. **Rotation-count cost model.** The cost is the realised rotation count and ignores the gate-synthesis cost of individual rotations and hardware connectivity; the antithetic schedule’s seam rotations could be merged to reduce the count further, which we do not exploit so that its budget matches first-order exactly.
5. **Classical simulation; solo author.** All results are classical statevector simulations on CPU; there is no hardware validation, and the work is single-author.

Conclusion. The free parameter in first-order Trotterisation is not the ordering but the schedule. A single ordering is provably impotent—it only re-signs the leading commutator error—whereas a step-alternating antithetic schedule cancels that error and converges at second order at the identical gate budget, reaching target fidelities that first-order Trotter cannot. The commutator graph certifies the impotence and structures the schedule, and a small learned router recovers near-oracle gate efficiency by folding only the instances that need it. The natural next steps are larger systems with scalable references, real mapped molecular Hamiltonians, higher-order folded schedules, and a graph neural policy trained directly on a differentiable matched-cost objective. All experiments are reproducible on commodity hardware; runtime and memory are reported for each benchmark.

4 Methods

This section gives the formal development. The Pauli algebra (Sec. 4.1) yields the exact matrix-free simulator and the commutator error form; Sec. 4.2 states and proves cost invariance, the ordering-impotence theorem, the antithetic-cancellation theorem, the clique exactness, and the listing-order invariance of the learned router. All theorems are stated under explicitly named hypotheses with complete proofs.

4.1 Pauli algebra and the exact statevector simulator

A Pauli string $P \in \{I, X, Y, Z\}^{\otimes n}$ acts on $|\psi\rangle \in \mathbb{C}^{2^n}$. We write $\{A, B\} = AB + BA$ and $[A, B] = AB - BA$. The single-qubit Paulis satisfy $X^2 = Y^2 = Z^2 = I$ and $\{X, Y\} = \{Y, Z\} = \{Z, X\} = 0$. We record the structural facts that make an exact matrix-free simulator possible, each with a complete proof.

Lemma 1 (Commutation is an $O(n)$ parity test). *Let $P = \bigotimes_{q=1}^n p_q$ and $Q = \bigotimes_{q=1}^n q_q$ be Pauli strings. Call position q anticommuting if $p_q \neq q_q$ and neither is I , and let $k = k(P, Q)$ be the number of anticommuting positions. Then $PQ = (-1)^k QP$, so P and Q commute iff k is even. The test runs in $O(n)$ time and forms no $2^n \times 2^n$ matrix.*

Proof. At each position either $p_q = q_q$ or one is I (then $p_q q_q = q_q p_q$), or $p_q \neq q_q$ and neither is I (then $p_q q_q = -q_q p_q$ by $\{X, Y\} = \{Y, Z\} = \{Z, X\} = 0$). So $p_q q_q = (-1)^{\epsilon_q} q_q p_q$ with $\epsilon_q = 1$ exactly on anticommuting positions. As the tensor product factorises across qubits, $PQ = \bigotimes_q p_q q_q = (\prod_q (-1)^{\epsilon_q}) \bigotimes_q q_q p_q = (-1)^k QP$, $k = \sum_q \epsilon_q$. Hence $PQ = QP$ iff k even. $\square$

Lemma 2 (Pauli rotation as a cosine-sine pair). *For a non-identity Pauli string P ($P^2 = I$) and $\theta \in \mathbb{R}$,*

$$e^{-i\theta P} = \cos \theta I - i \sin \theta P, \quad e^{-i\theta P} |\psi\rangle = \cos \theta |\psi\rangle - i \sin \theta P |\psi\rangle. \quad (5)$$

The right-hand side is one sparse application of P plus a scaled vector add, computable in $O(n 2^n)$ time without forming the $2^n \times 2^n$ matrix.

Proof. Since $P^{2m} = I$ and $P^{2m+1} = P$, regrouping the absolutely convergent series of $e^{-i\theta P}$ by parity gives $I \sum_m (-1)^m \theta^{2m} / (2m)! - iP \sum_m (-1)^m \theta^{2m+1} / (2m+1)! = \cos \theta I - i \sin \theta P$. A single Pauli string maps each basis state to one basis state with a unit-modulus phase, so $P |\psi\rangle$ costs $O(n 2^n)$. $\square$

Before any fidelity number is produced, Eq. (5) is cross-checked against dense `scipy.linalg.expm` on random small Paulis to tolerance 10^{-9} (the `Pauli_algebra_verified` flag). The product of two Pauli strings is itself a phased Pauli, $PQ = \omega R$ with $\omega \in \{\pm 1, \pm i\}$ and R a Pauli string, computed in $O(n)$ (the `Pauli_product` kernel); this is the algebraic basis of the commutator error form below.

4.2 Cost invariance, ordering impotence, and antithetic cancellation

The commutator graph $\mathcal{G}(H)$ has one node per term and an edge $\{j, k\}$ for every anticommuting pair (Lemma 1). A *schedule* assigns to each of the r Trotter steps a permutation of $\{1, \dots, L\}$; a first-order schedule uses the same permutation π every step. We first record that the gate budget is a schedule invariant for the schedules we compare at matched cost.

Proposition 3 (Rotation-count invariance). *A first-order schedule with any ordering π , and the antithetic schedule that alternates π with its reverse, each apply exactly $L \cdot r$ single-Pauli rotations for r steps. Hence their rotation count is independent of the ordering and of the alternation.*

Proof. One first-order step is an ordered product of exactly one factor per term, L rotations; r steps give $L \cdot r$. The antithetic schedule applies, at each step, a permutation of the same L factors (forward on even steps, reversed on odd), so each step is again L rotations and the total is $L \cdot r$; reversing a step's order permutes its factors without adding or removing any. $\square$

The next lemma pins the exact value of the Pauli commutators that populate the error form.

Lemma 4 (Pauli commutator value). *Let $h_j = c_j P_j$, $h_k = c_k P_k$ with real c_j, c_k . If P_j, P_k commute then $[h_j, h_k] = 0$. If they anticommute then $[h_j, h_k] = 2c_j c_k P_j P_k = 2c_j c_k \omega_{jk} R_{jk}$ with $\omega_{jk} \in \{\pm i\}$ and R_{jk} a Hermitian Pauli string, so $\|[h_j, h_k]\|_{\text{HS}} = 2|c_j c_k| 2^{n/2}$ and the operator norm is $2|c_j c_k|$.*

Proof. By Lemma 1, $P_k P_j = (-1)^{k(P_j, P_k)} P_j P_k$. If they commute, $[h_j, h_k] = c_j c_k (P_j P_k - P_k P_j) = 0$. If they anticommute, $P_k P_j = -P_j P_k$, so $[h_j, h_k] = 2c_j c_k P_j P_k$. The product $P_j P_k = \omega_{jk} R_{jk}$ is a phased Pauli (Sec. 4.1); since the commutator of Hermitians is anti-Hermitian and R_{jk} is Hermitian, $\omega_{jk} \in \{\pm i\}$. Every Pauli string is unitary with $\|R_{jk}\| = 1$ and $\|R_{jk}\|_{\text{HS}}^2 = \text{Tr}(R_{jk}^2) = 2^n$, giving the stated norms. $\square$

Definition 1 (Commutator error form). *For an ordering π , the leading error operator is $E_\pi = \sum_{a < b} [h_{\pi(a)}, h_{\pi(b)}]$. By Lemma 4 it equals the edge sum Eq. (4), a signed sum of phased Pauli strings indexed by the edges of $\mathcal{G}(H)$, where $\varepsilon_{jk}(\pi) = +1$ if j precedes k in π and -1 otherwise. It is assembled in $O(L^2 n)$ time without forming a $2^n \times 2^n$ matrix, and is validated against the dense pair-commutator sum to 10^{-9} (the `error_form_verified` flag).*

Theorem 5 (Ordering impotence). *Let $H = \sum_j c_j P_j$ and let E_π be its commutator error form. Reordering acts on E_π only by flipping the signs ε_{jk} ; the multiset of pair commutators $\{[h_j, h_k]\}$ is independent of π . Consequently, grouping the edge sum by Pauli string,*

$$\|E_\pi\|_{\text{HS}}^2 = 2^n \sum_R \left(\sum_{\{j,k\}: R_{jk}=R} \varepsilon_{jk}(\pi) 2c_j c_k \hat{\omega}_{jk} \right)^2,$$

where $\hat{\omega}_{jk} = \omega_{jk}/i \in \{\pm 1\}$. *If every edge maps to a distinct Pauli string R (no collisions), then $\|E_\pi\|_{\text{HS}}^2 = 2^n \sum_{\{j,k\}} (2c_j c_k)^2$ is independent of π . In general only the colliding edges contribute an ordering-dependent term, so $\|E_\pi\|_{\text{HS}}$ is bounded below by the ordering-invariant contribution of the non-colliding edges for every π .*

Proof. For an unordered pair $\{j, k\}$, listing j before k contributes $[h_j, h_k]$ and listing k before j contributes $[h_k, h_j] = -[h_j, h_k]$; hence reordering replaces the contribution by its negative, i.e. multiplies it by $\varepsilon_{jk}(\pi)$, and never removes or adds a pair. By Lemma 4 each anticommuting edge contributes $2c_j c_k \omega_{jk} R_{jk} = i 2c_j c_k \hat{\omega}_{jk} R_{jk}$. Distinct Pauli strings are Hilbert–Schmidt orthogonal with squared norm 2^n , so $\|E_\pi\|_{\text{HS}}^2 = 2^n \sum_R |\alpha_R(\pi)|^2$ with $\alpha_R(\pi) = i \sum_{\{j,k\}: R_{jk}=R} \varepsilon_{jk}(\pi) 2c_j c_k \hat{\omega}_{jk}$, which is the displayed expression. If each R has a single contributing edge then $|\alpha_R| = 2|c_j c_k|$ regardless of the sign ε_{jk} , so the sum is π -independent. Edges that are alone on their Pauli string therefore contribute a fixed amount to $\|E_\pi\|_{\text{HS}}^2$ for every π ; the remaining (colliding) edges are the only source of ordering dependence, and their contribution is non-negative, giving the stated lower bound. $\square$

Theorem 5 is the precise content of “ordering is within noise”: on the transverse-field Ising and Heisenberg families every anticommuting edge maps to a distinct Pauli string (Table 1, ordering-invariant fraction 1.00), so $\|E_\pi\|_{\text{HS}}$ is *exactly* the same for every ordering, and on the molecular-like family only a small colliding fraction is reducible. We now show that enlarging from orderings to schedules escapes this floor.

Theorem 6 (Antithetic cancellation and the free order upgrade). *Fix an ordering π and step size $\Delta = t/r$ with r even, and let $S_\pi(\Delta) = \prod_{a=1}^L e^{-i\Delta h_{\pi(a)}}$ and $S_{\pi R}(\Delta) = \prod_{a=L}^1 e^{-i\Delta h_{\pi(a)}}$ be the forward and reversed steps. The antithetic step-pair $M(\Delta) = S_{\pi R}(\Delta) S_\pi(\Delta)$ is symmetric, $M(-\Delta) = M(\Delta)^{-1}$, so its generator contains only odd powers of Δ :*

$$M(\Delta) = \exp(-2i\Delta H + O(\Delta^3)),$$

and the leading $O(\Delta^2)$ commutator error $-\frac{\Delta^2}{2} E_\pi$ cancels. Consequently the antithetic schedule $(S_{\pi R}(\Delta) S_\pi(\Delta))^{r/2}$ approximates e^{-itH} with additive error $O(t^3/r^2)$, while applying exactly $L \cdot r$ rotations—the identical budget of the first-order schedule (Proposition 3).

Proof. Write the factors of $M(\Delta)$ left to right: $e^{-i\Delta h_{\pi(L)}}, \dots, e^{-i\Delta h_{\pi(1)}}, e^{-i\Delta h_{\pi(1)}}, \dots, e^{-i\Delta h_{\pi(L)}}$, a palindromic sequence. Reversing the order of the factors and replacing each by its inverse

(i.e. $\Delta \mapsto -\Delta$) returns the same product read backwards, so $M(-\Delta) = M(\Delta)^{-1}$. Writing $M(\Delta) = \exp(\Omega(\Delta))$ with $\Omega(\Delta) = \sum_{m \geq 1} \Omega_m \Delta^m$ (Baker–Campbell–Hausdorff, convergent for small Δ), the relation $M(-\Delta) = M(\Delta)^{-1}$ reads $\exp(\Omega(-\Delta)) = \exp(-\Omega(\Delta))$, i.e. $\Omega(-\Delta) = -\Omega(\Delta)$, so all even-order terms Ω_{2m} vanish; in particular the Δ^2 term, which for the ordered product $\prod_a e^{-i\Delta h_a}$ equals $-\frac{1}{2} \sum_{a < b} [h_a, h_b]$ before symmetrisation, is removed. Each term h_j appears twice in M , once in each half, so $\Omega_1 = -2iH$ and $M(\Delta) = \exp(-2i\Delta H + O(\Delta^3))$. Thus one antithetic pair is a second-order approximant of $e^{-2i\Delta H}$; composing $r/2$ pairs with $\Delta = t/r$ telescopes (as in the standard product-formula bound) to additive error $r/2 \cdot O(\Delta^3) = O(t^3/r^2)$. The pair applies $2L$ rotations and there are $r/2$ pairs, giving $L \cdot r$ rotations in total. $\square$

Proposition 7 (Clique exactness). *A proper colouring of $\mathcal{G}(H)$ partitions the terms into cliques of pairwise commuting Pauli terms. Laying a clique C out contiguously in the schedule, its rotations satisfy $\prod_{j \in C} e^{-i\Delta h_j} = \exp(-i\Delta \sum_{j \in C} h_j)$ exactly for any internal order, so no first-order error arises among the members of a clique; the schedule error depends only on inter-clique commutators. The number of non-commuting units is reduced from L to the number of cliques, which is at least the chromatic number $\chi(\mathcal{G})$.*

Proof. In a proper colouring no two adjacent vertices share a colour, and adjacency in $\mathcal{G}(H)$ is anticommutation, so same-colour terms commute. For pairwise commuting $\{h_j\}_{j \in C}$ the exponentials factor exactly, $\prod_{j \in C} e^{-i\Delta h_j} = \exp(-i\Delta \sum_{j \in C} h_j)$, independent of internal order; hence by Lemma 4 no intra-clique pair contributes to E_π , and only inter-clique edges remain. $\square$

For low-chromatic Hamiltonians (e.g. Heisenberg, where the bond operators colour into a few commuting classes) a clique-ordered first-order schedule is therefore near-exact at a single step, which is why folding adds only overhead there (Sec. 2.4).

4.3 The learned router

Ten listing-order-invariant features of $\mathcal{G}(H)$ are computed per instance: term count; anticommutation (edge) density; mean and maximum node degree; number of greedy colour cliques; largest-clique fraction; coefficient coefficient-of-variation; mean Pauli weight; and two commutator-error-form features—the colliding fraction and the ordering-invariant fraction (Definition 1). A standardised `LogisticRegression` classifier [16] maps these to the label “which fixed schedule reaches the target fidelity at the fewest rotations on this instance,” trained leave-one-instance-out (each test instance scored by a classifier trained only on the other $120 - 1$ instances), falling back to the symmetric schedule when training labels are single-class.

Proposition 8 (Listing-order invariance of the router). *Let $\sigma \in S_L$ relabel the terms of H . Each of the ten features is a function of the unordered weighted graph $\mathcal{G}(H)$ and is invariant under σ . Consequently the router’s predicted schedule depends only on the isomorphism class of $\mathcal{G}(H)$, not on how the terms were listed.*

Proof. Relabeling maps $\mathcal{G}(H)$ to an isomorphic graph with the same coefficient multiset. Vertex/edge counts, the degree multiset, the greedy-colouring clique counts (under a deterministic degree-driven tie-break), the coefficient coefficient-of-variation, and the mean Pauli weight are graph isomorphism invariants. The colliding and ordering-invariant fractions count, over unordered anticommuting pairs, how the products $P_j P_k$ coincide—a symmetric function of the unordered term multiset (Definition 1). Each feature is therefore invariant under σ , so the standardised classifier’s prediction is too. $\square$

4.4 Hamiltonian families and protocol

Three families are generated as $\{(c_j, P_j)\}$ term lists: transverse-field Ising $H = -J \sum_i Z_i Z_{i+1} - h \sum_i X_i$; Heisenberg $H = \sum_i (J_x X_i X_{i+1} + J_y Y_i Y_{i+1} + J_z Z_i Z_{i+1})$; and *molecular-like*, random local 2-body Pauli terms with seeded Gaussian coefficients (a tractable stand-in for the dense,

irregular anticommutation structure of Jordan–Wigner / Bravyi–Kitaev mapped electronic-structure Hamiltonians [17, 18]). The reported-scale configuration (`configs/full.yaml`) uses sizes $n \in \{4, 5, 6, 7, 8\}$, eight seeded instances per (family, size), evolution time $t = 1.0$, a reference budget of $r = 6$ steps, and a step grid $\{1, 2, 4, 6, 8, 12, 16, 24\}$ for the convergence and frontier curves, giving 120 instances. For each instance and schedule we evaluate the product formula, score fidelity and infidelity against the exact reference, and record the realised rotation count. From the (rotation count, fidelity) curves we compute the rotations to reach each target fidelity (the matched-cost efficiency metric) and the fitted convergence rate; ordering impotence is measured by sampling 48 orderings per instance and forming the exact leading-error spectral norm. Per-schedule means carry 95% confidence intervals $1.96 \hat{\sigma} / \sqrt{N}$.

4.5 Reproducibility, software, and provenance

The pipeline is a CPU-only Python package (`topoham`) depending on NumPy [19], SciPy [20], NetworkX [21], scikit-learn [16], and Matplotlib. The single source of truth is `results/summary.json`, written by the runner with full provenance (seed, command, platform, package versions, runtime, peak memory) and two integrity flags: `pauli_algebra_verified` (the fast kernel matches dense `expm` to 10^{-9}) and `error_form_verified` (the matrix-free commutator error form matches the dense pair-commutator sum to 10^{-9}). On the run reported here (seed 0, Python 3.13, macOS arm64), the experiment completed in ≈ 49 s wall-clock with ≈ 496 MB peak memory; both flags were `true`. The macros, tables, and figures in this manuscript are generated directly from `summary.json` by `make_tables.py` and `make_figures.py` and \input'd here, so no number is entered by hand. All experiments are reproducible on commodity hardware; runtime and memory are reported for each benchmark.

Data availability

This study uses no external datasets. All Hamiltonians are generated synthetically from fixed random seeds by the accompanying code, and all numerical values, tables, and figures are produced by that code and stored in `results/summary.json`.

Code availability

The code that implements the method and regenerates every figure, table, and reported number accompanies this manuscript in `submission/code` as an installable package (`pip install .`) and will be released in a public repository upon publication.

Competing interests

The author declares no competing interests.

Author contributions

M.H. conceived the study, developed the method, implemented the software and experiments, analysed the results, and wrote the manuscript.

Funding

The author received no specific funding for this work.

Acknowledgements

The author thanks the maintainers of the open-source scientific Python ecosystem (NumPy, SciPy, NetworkX, scikit-learn, and Matplotlib), on which this work relies.

Supplementary Information

This Supplementary Information is a self-contained, from-scratch derivation of every concept the accompanying code (`submission/code`) implements, mapped to its source file. It assumes linear algebra and basic probability but not quantum computing, product formulas, the commutator graph, or the schedules used here.

S1 The problem in one paragraph

We want to simulate $e^{-iHt}|\psi_0\rangle$. On a quantum computer we approximate e^{-iHt} by a *product formula* that chains together easy one-term rotations. The cost of a first-order schedule is one rotation per term per step, and this cost does not depend on the order in which the rotations are applied—nor on whether that order *varies between steps*. The error does. The scientific question is which use of this freedom reduces the error: we show a single fixed order cannot (Sec. S5), while a step-alternating *schedule* can, for free (Sec. S6).

S2 Hamiltonians as Pauli sums

Source: `src/topoham/hamiltonians.py`. Every Hamiltonian on n qubits is a real combination of Pauli strings, $H = \sum_j c_j P_j$, each P_j a length- n word over $\{I, X, Y, Z\}$. Three families are generated as term lists: transverse-field Ising, Heisenberg, and a molecular-like family of random local two-body terms with seeded Gaussian coefficients—spanning sparse-and-fast-forwardable to dense-and-stubborn anticommutation structure.

S3 Pauli algebra: three facts that make exact simulation cheap

Source: `src/topoham/pauli.py`. (A) Two Pauli strings commute iff they anticommute on an even number of qubits—an $O(n)$ parity test (Lemma 1). (B) A non-identity Pauli is an involution, so $e^{-i\theta P}|\psi\rangle = \cos\theta|\psi\rangle - i\sin\theta(P|\psi\rangle)$ is a single sparse application of P plus a scaled add (Lemma 2); this is the inner kernel of the simulator, and the in-browser demo applies the same identity. (C) The product of two Pauli strings is a phased Pauli, $PQ = \omega R$ with $\omega \in \{\pm 1, \pm i\}$ (`pauli_product`)—the algebra behind the commutator error form. Before any fidelity is produced, the fast kernel is checked against dense `expm` to 10^{-9} (`pauli_algebra_verified`).

S4 The commutator graph and the error form

Source: `src/topoham/commutator_graph.py`, `src/topoham/error_form.py`. The graph $\mathcal{G}(H)$ has one node per term and an edge between anticommuting pairs. Its proper colouring partitions the terms into mutually commuting cliques (Proposition 7), each exponentiable exactly and simultaneously. The *commutator error form* (Definition 1) is the exact leading error operator $E_\pi = \sum_{a < b} [h_{\pi(a)}, h_{\pi(b)}]$, built matrix-free as a signed sum of phased Pauli strings over the edges. The module also computes the ten listing-order-invariant features the learned router reads, including the colliding and ordering-invariant fractions, and validates the matrix-free form against the dense pair-commutator sum to 10^{-9} (`error_form_verified`).

S5 Why ordering is impotent (the theorem, derived)

Source: Methods, Sec. 4.2. For Pauli terms the only ordering-dependent feature of E_π is the sign ε_{jk} of each edge’s commutator: listing j before k gives $[h_j, h_k]$, the reverse gives its negative. Because distinct Pauli strings are Hilbert–Schmidt orthogonal, the squared HS norm of E_π is a sum over Pauli strings of squared coefficients; an edge alone on its Pauli string contributes a fixed amount regardless of its sign. Hence on families with no colliding edges $\|E_\pi\|_{\text{HS}}$ is exactly the same for every ordering, and in general only the colliding fraction is reducible (Theorem 5). Empirically the spectral norm of a random ordering is within a factor 1.085 of the per-instance minimum: ordering cannot escape the $O(t^2/r)$ rate.

S6 Why a schedule escapes it (the cancellation theorem)

Source: src/topoham/schedules.py. A *schedule* may apply a different order at each step. The *antithetic* schedule alternates an order π with its reverse π^R ; each adjacent step-pair $S_{\pi^R}(\Delta)S_\pi(\Delta)$ reads as a palindrome, so it is a symmetric formula whose generator has only odd powers of Δ , cancelling the $O(\Delta^2)$ commutator error (Theorem 6). The result is second-order accuracy, $O(t^3/r^2)$, at exactly $L \cdot r$ rotations—the same as first-order. The *symmetric* (Strang) schedule applies each term twice per step at half angle with seam merges, a lower-constant second-order fold at $\approx 2L \cdot r$ rotations. Both fold over the clique order so that intra-clique error is already absent (Proposition 7).

S7 Schedules, the matched-cost metric, and the learned router

Source: src/topoham/schedules.py, policies.py, env.py. A schedule is realised as a flat list of (term, step-fraction) rotations; its (seam-merged) length is the gate cost on which all schedules are compared. For each instance and schedule and each step count r in the grid, the environment builds the exact reference $e^{-iHt}|+\rangle^{\otimes n}$ and scores fidelity, and from the resulting (cost, fidelity) curve we read the rotations to a target fidelity. The learned router (`LearnedSchedulePolicy`) is a standardised logistic-regression classifier over the ten graph features that predicts the gate-optimal fixed schedule, trained leave-one-instance-out (Sec. 4.3); it returns the predicted schedule, a selector over human-readable schedules rather than a black box, and its decision is listing-order invariant (Proposition 8).

S8 Metrics, aggregation, and provenance

Source: metrics.py, runner.py, make_tables.py, make_figures.py. Per schedule we report the mean fidelity with a 95% confidence interval $1.96 \hat{\sigma}/\sqrt{N}$, the convergence rate (mean per-instance log-log slope of infidelity vs. r), and the rotations to each target fidelity, overall and per family. `runner.run` writes `results/summary.json`—the single source of truth—with full provenance and the two integrity flags. The manuscript `\input`’s the generated macros and table fragments and reads the figures from `code/figures/`, so nothing is typed by hand. An audit (`make audit`) checks that the headline macros are present, the integrity flags are set, and no forbidden superiority phrasing appears.

S9 How to reproduce everything

From `submission/code`:

```
pip install .           # installs the topoham package
make test               # Pauli kernel (1e-9), error-form (1e-9),
```



```
# antithetic 2nd-order, clique exactness
export KMP_DUPLICATE_LIB_OK=TRUE
topoham-reproduce --config configs/full.yaml # summary.json, tables, figures
```

The reported run (seed 0): 120 instances (3 families $\times$ 5 sizes $\times$ 8 seeded draws), time $t = 1.0$, reference budget $r = 6$, step grid $\{1, 2, 4, 6, 8, 12, 16, 24\}$. Headline: the antithetic schedule reaches fidelity 0.99 at 101 rotations against 152 for the best first-order ordering (a $1.62\times$ reduction) and 0.999 at 203 rotations where first-order Trotter does not reach it, lifting the convergence rate from 1.97 to 4.07 at the identical gate budget.

References

- [1] Richard P. Feynman. Simulating physics with computers. *International Journal of Theoretical Physics*, 21:467–488, 1982. doi: 10.1007/BF02650179.
- [2] Seth Lloyd. Universal quantum simulators. *Science*, 273(5278):1073–1078, 1996. doi: 10.1126/science.273.5278.1073.
- [3] Hale F. Trotter. On the product of semi-groups of operators. *Proceedings of the American Mathematical Society*, 10(4):545–551, 1959. doi: 10.1090/S0002-9939-1959-0108732-6.
- [4] Masuo Suzuki. General theory of fractal path integrals with applications to many-body theories and statistical physics. *Journal of Mathematical Physics*, 32(2):400–407, 1991. doi: 10.1063/1.529425.
- [5] Andrew M. Childs, Yuan Su, Minh C. Tran, Nathan Wiebe, and Shuchen Zhu. Theory of trotter error with commutator scaling. *Physical Review X*, 11(1):011020, 2021. doi: 10.1103/PhysRevX.11.011020.
- [6] Andrew M. Childs, Dmitri Maslov, Yunseong Nam, Neil J. Ross, and Yuan Su. Toward the first quantum simulation with quantum speedup. *Proceedings of the National Academy of Sciences*, 115(38):9456–9461, 2018. doi: 10.1073/pnas.1801723115.
- [7] Naomichi Hatano and Masuo Suzuki. Finding exponential product formulas of higher orders. *Lecture Notes in Physics*, 679:37–68, 2005. doi: 10.1007/11526216_2.
- [8] Andrew M. Childs, Aaron Ostrander, and Yuan Su. Faster quantum simulation by randomization. *Quantum*, 3:182, 2019. doi: 10.22331/q-2019-09-02-182.
- [9] Earl Campbell. Random compiler for fast Hamiltonian simulation. *Physical Review Letters*, 123(7):070503, 2019. doi: 10.1103/PhysRevLett.123.070503.
- [10] David Poulin, Matthew B. Hastings, Dave Wecker, Nathan Wiebe, Andrew C. Doherty, and Matthias Troyer. The trotter step size required for accurate quantum simulation of quantum chemistry. *Quantum Information & Computation*, 15(5–6):361–384, 2015.
- [11] Matthew B. Hastings, Dave Wecker, Bela Bauer, and Matthias Troyer. Improving quantum algorithms for quantum chemistry. *Quantum Information & Computation*, 15(1–2):1–21, 2015.
- [12] Vladyslav Verteletskyi, Tzu-Ching Yen, and Artur F. Izmaylov. Measurement optimization in the variational quantum eigensolver using a minimum clique cover. *The Journal of Chemical Physics*, 152(12):124114, 2020. doi: 10.1063/1.5141458.

- [13] Pranav Gokhale, Olivia Angiuli, Yongshan Ding, Kaiwen Gui, Teague Tomesh, Martin Suchara, Margaret Martonosi, and Frederic T. Chong. $O(N^3)$ measurement cost for variational quantum eigensolver on molecular hamiltonians. *IEEE Transactions on Quantum Engineering*, 1:1–24, 2020. doi: 10.1109/TQE.2020.3035814.
- [14] Ophelia Crawford, Billy van Straaten, Daochen Wang, Thomas Parks, Earl Campbell, and Stephen Brierley. Efficient quantum measurement of pauli operators in the presence of finite sampling error. *Quantum*, 5:385, 2021. doi: 10.22331/q-2021-01-20-385.
- [15] Andrew Tranter, Peter J. Love, Florian Mintert, and Nathan Wiebe. Ordering of trotterization: Impact on errors in quantum simulation of electronic structure. *Entropy*, 21(12):1218, 2019. doi: 10.3390/e21121218.
- [16] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, et al. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [17] Sergey B. Bravyi and Alexei Yu. Kitaev. Fermionic quantum computation. *Annals of Physics*, 298(1):210–226, 2002. doi: 10.1006/aphy.2002.6254.
- [18] James D. Whitfield, Jacob Biamonte, and Alán Aspuru-Guzik. Simulation of electronic structure hamiltonians using quantum computers. *Molecular Physics*, 109(5):735–750, 2011. doi: 10.1080/00268976.2011.552441.
- [19] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, et al. Array programming with numpy. *Nature*, 585:357–362, 2020. doi: 10.1038/s41586-020-2649-2.
- [20] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. Scipy 1.0: fundamental algorithms for scientific computing in python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.
- [21] Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. Exploring network structure, dynamics, and function using networkx. In *Proceedings of the 7th Python in Science Conference*, pages 11–15, 2008.